CS & IT ENGINEERING

Computer Organization Architecture

Cache Organization

DPP- 02

Discussion Notes



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[MCQ]

2^5 B

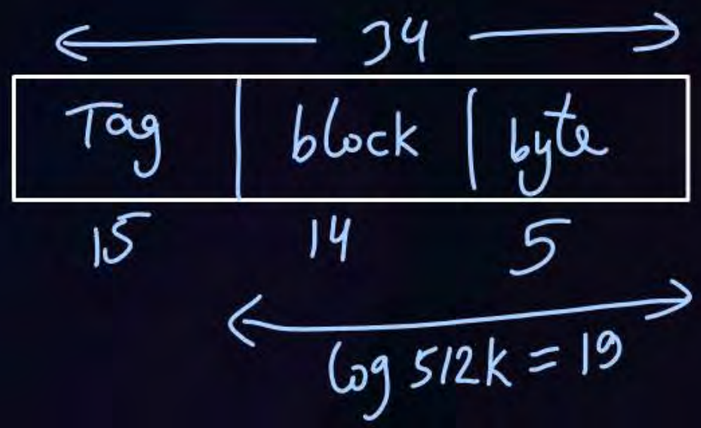
#Q. Consider a 512KB direct mapped cache with block size of 32 bytes. The main memory address is of 34-bits. The size of index and tag in bits are?

A 15, 14

B 14, 15

C 19, 14

D 14, 19

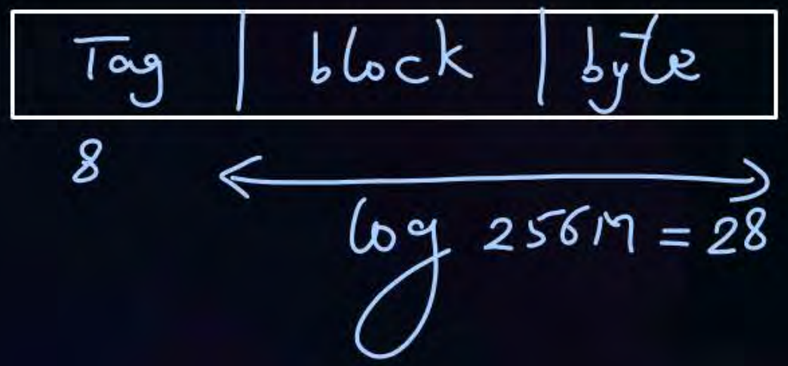


no. of blocks in cache $= \frac{512kB}{32B} = \frac{2^{19}}{2^5} = 2^{14}$

[NAT]

Ans = 64

#Q. Consider a direct mapped cache of size 256MB. Cache controller maintains 8-bits tag for each block in cache. The maximum size of main memory (byte addressable) supported in the system is _____GB?



mm add. = 36 bits

mm size = 2^{36} bytes

= 64 GB

[NAT]

Ans = 256

#Q. The size of memory required at cache controller to store metadata is 2KBytes. The metadata includes tag bits, 1 modified bit and 1 valid bit. The cache contains 1K blocks of 16 bytes each and organized as direct mapped. The size of main memory is___ Mbytes?

mm add.

Tag	Block	byte
14	10	4

mm add. = 28 bits

mm size = 2^{28} bytes

= 256 MB

Tag directory size = no. of blocks * (Tag+2) bits

2K * 8 bits = 1K * (Tag +2)

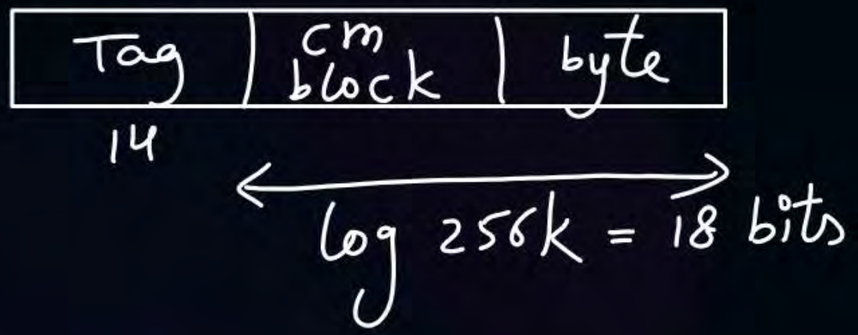
Tag = 14 bits

[NAT]

Ans = 32

#Q. Consider a direct mapped cache of size 256KB. The CPU generates x- bit addresses. The number of tag bits in main memory address are 14 bits then value of x is _____?

mm add.



mm add. = 14 + 18

= 32 bits

[MCQ]

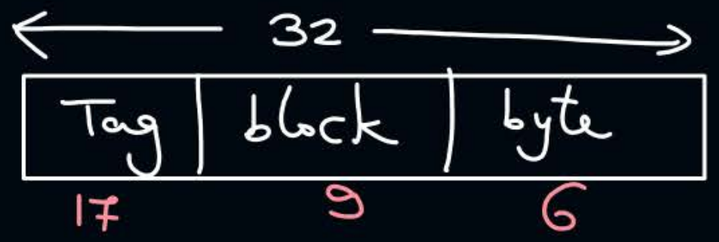
#Q. Assume a computer has 32-bit addresses. Each block stores 64 bytes. A direct-mapped cache has 512 blocks. Match the block (line) of the cache (in decimal) we look for each of the given hexadecimal addresses in the table?

A. 1A2BC012: 256, FFFF00FF: 3, 12345678: 345, C109D532: 340

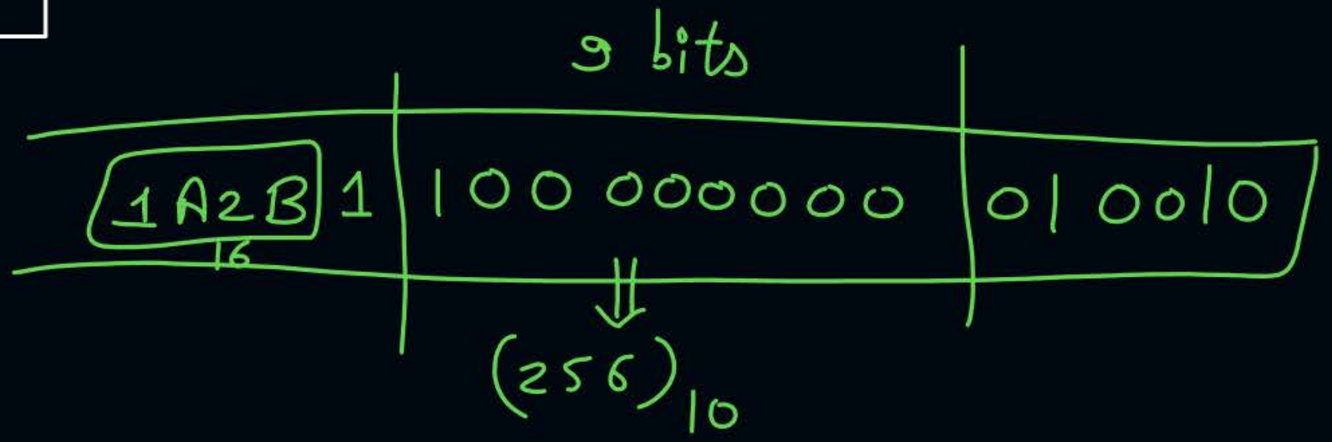
B. 1A2BC012: 512, FFFF00FF: 7, 12345678: 243, C109D532: 320

C. 1A2BC012: 128, FFFF00FF: 5, 12345678: 345, C109D532: 420

D. 1A2BC012: 255, FFFF00FF: 1, 12345678: 247 , C109D532: 240



1.) 1A2BC012



2) FFFF00FF $(00000000\,11)_2 = (3)_{10}$

3) $(1234\,5678)_{16}$ $(101\,011001)_2 = (345)_{10}$

4) $(C109D532)_{16}$ $(101010100)_2 = (340)_{10}$

[NAT]

Ans = 4

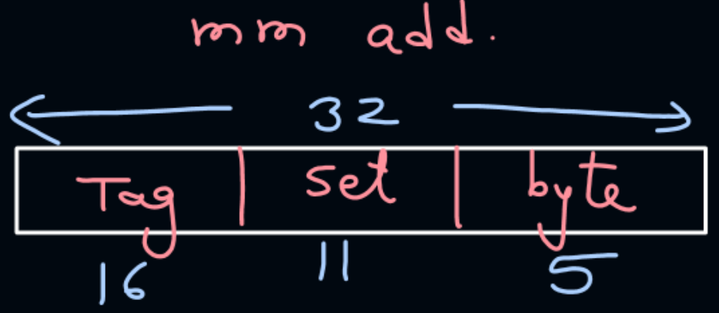
#Q. Consider a cache with 2^{13} blocks of size 32 Bytes each. The CPU generates addresses of 32-bits. The cache controller stores 1 valid bit, 1 modified bit and tag-bits for each metadata entry. The cache controller has a maximum memory of 18Kbytes to store the metadata. The cache is organized as k-way set associative. Maximum value of k to utilize the cache controller memory in optimized manner is ___?

2^5

Tag directory size = 18KB = $2^{13} * (Tag + 2)$ bits

18K * 8 bits = $2^{13} * (Tag + 2)$ bits

Tag = 16 bits



$$\text{no. of sets} = 2^{11} = \frac{\text{no. of blocks in cm}}{k}$$

$$2^{11} = \frac{2^{13}}{k}$$

$$k = \frac{2^{13}}{2^{11}} = 2^2 = \underline{\underline{4}}$$

[NAT]

Ans = 97.8 %

#Q. Consider a direct mapped write back data cache of size 2KB with the block size of 128 bytes. The cache is considered to be empty initially. The byte addressable main memory has size 1Mbytes. Further consider that there is an array A[35][20] with each element occupies 4 bytes. The base address of array is $(1A300)_{16}$. The array is accessed 3 times. And between the accesses, there is no any data cache changes happen. Hit ratio (correct upto 1 decimal place) of cache for this array access is ___%?

$$\text{no. of blocks in cache} = \frac{2\text{kB}}{128\text{B}} = \frac{2^{11}}{2^{7}} = 2^{4} = 16$$

Array size = 35 * 20 = 700 elements

= 700 * 4 = 2800 Bytes

no. of blocks needed to store array = $\left\lceil \frac{2800\,B}{128\,B} \right\rceil$ = 22 blocks

overlapping blocks = 22 − 16 = 6

	no. of miss
1st access	22
2nd access	6 * 2 = 12
3rd "	6 * 2 = 12
Total	46

Total no. of mem. reference = 700 * 3
= 2100

No. of hits = 2100 − 46 = 2054

Hit ratio = $\frac{2054}{2100}$ = 0.978 = 97.8 %

THANK - YOU